

Haoran Feng

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EDUCATION

Sep 2025 - Jun 2027 (Expected)	CUHK-Shenzhen	M.S., Integrated Circuits and Systems
Sep 2021 - Jun 2025	Shandong University (985)	B.Eng., Electronic Science / Computer Science

Honors: Academic Scholarship (Top 20%); Special Talent Scholarship in Innovation, Shandong University

INTERNSHIP EXPERIENCE

Apr 2026 - Present	Ubiquant Investment - AI Lab	LLM and HPC Intern
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• Training Kernel Support

- FA3 deterministic SWA backward:** Used Nsys/NCU factorization to attribute 99.67% of deterministic overhead to the dQ semaphore chain. Implemented contributor-relative tickets aligned with the reverse scheduler in the fused kernel, with no new kernel/workspace/D2H and a short-sequence fallback. Cut packed backward from 6.755 to 1.699 ms (3.98x), achieved 1.59-5.37x at 2K-12K, and passed 3,601/3,601 regressions plus 1,000 bitwise repeats.
- Sink FC1 GEMM and communication contention:** Corrected FC1 to BF16 8192x3072x3072 and isolated cross-stream communication contention rather than a slow GEMM. Built a 4xH200 dual-stream proxy, sweeping NCCL CTA budgets and DeepGEMM SM budgets (121 coarse, 89 fine) with Nsys grid/overlap checks. In the AllGather proxy, 12 CTA + DG120 reduced joint span by 12.80%, including 7.58% independent gain from compute throttling; AllToAll gains were attributed to communication parallelism.

• Inference Kernel Support

- Deterministic MoE Router GEMM under CUDA Graph:** Integrated DeepGEMM, Triton Full-K, and persistent backends in vLLM; owned the tensor-signature selector, pre-capture numeric/graph preflight, cache/fallback, workspace lifecycle, and path tests. Replay freezes the backend decision. Reduced median latency of 48 router GEMMs by 73.39%, passed 135/135 kernel and 20/20 model bitwise checks, and improved prefix-cache-hit request throughput by 3.81%/4.36% on TP1/TP2.
- Asynchronous OE state operator:** Kept recent-token history and oe_input_ids construction on GPU with Triton fused hashing, fixing cross-step state for prefill/decode/mixed batches/recovery/slot reuse/reorder. Passed 28/28 TP1 cases with +4.99% throughput; the TP>1 async-unfused batch-invariant path reached 84/84 with zero mismatches and +4.6%/+3.0% decode throughput on TP2/TP4.

• RL Training-Inference Consistency and Rollout Reliability

- R3 Router Replay and multi-GPU observability:** Built token/layer/top-k route capture, transport, training-side replay, response-mask alignment, and mismatch/f-tau2/KL diagnostics between vLLM and Megatron. On 8xH200/Qwen3-30B-A3B, reduced route mismatch from 17-19% to 0, f-tau2 by 36-145x, and KL by 4-7x; extended to a monitored 128-GPU, 200-step rollout with aligned key metrics.
- FlashInfer CUDA Graph hang:** Reproduced a vLLM TP=2 FULL CUDA Graph rollout hang on two H200 GPUs. TP/eager/graph/fused controls traced it to FTZ misclassifying a Lamport -0.0 sentinel in fused AllReduce + RMSNorm. Backported the upstream 0x80000000 bitwise fix; repeated model graph on/off replays had no hang or timeout.

Jan 2026 - Apr 2026

Moore Threads

Operator & Compiler Optimization Intern

- TensorFlow MUSA Extension, graph optimization, and reliability:** Integrated muDNN GELU, repaired the fusion pipeline, and optimized hot paths; all 11 GELUs fused and representative-shape latency fell 36.6%. Reworked HostMemory after tracing shape tensors entering the device path, lifting success from about 30% at 500 iterations to 100% at 1,000 and passing 40K/400K/800K stress runs. Reduced Logical_Or from 21.2 to 10.7 us and raised end-to-end throughput from 8187.48 to 8284.65.

OPEN-SOURCE CONTRIBUTIONS

Aug - Sep 2026

Merged External Contributions

- 5 merged external PRs (XFDG):** Mooncake [#3601/#3604/#3660/#3726](#) (Merged), covering storage durability, RDMA recovery, memory-unregister lifecycle, and queue error propagation; Mirage [#755](#) (Merged) fixes argmax padding rows.

PROJECTS AND RESEARCH

Dec 2025 - Present

High-Performance Quantized LLM Runtime with PyTorch Extensions

- **Quantized runtime, kernel debugging, and TP2 deployment:** Added configurable FlashAttention-2/4, GQA KV-head expansion, soft-cap masking, SDPA fallback, and multi-GPU builds to a W4A16/W8A8/FP8 runtime. Used Compute Sanitizer to fix a tail-group out-of-bounds access and implemented packed-AWQ TP=2 sharding with NCCL all-reduce. All four Sanitizer modes passed representative shapes; two-GPU Qwen2.5-Coder-32B W4A16 reduced checkpoint/peak memory by 70.5%/66.8% and increased end-to-end output throughput by 76.8%.

Jan 2026 - Present

Token-Weighted CoT Distillation | AAI 2027 Submission | [Code](#)

- **Method and results:** Trained and tuned a three-stage structure-recovery, token-weighted supervision, and preference-optimization pipeline; token importance was estimated from reference-answer likelihood drops after teacher-rationale perturbations. Completed LoRA/DPO training and vLLM TP=4 evaluation. Qwen2.5-7B-Instruct reached 94.01%/94.00% on GSM8K/SVAMP, +5.51/+10.10 points over the strongest baselines; manuscript submitted to AAI 2027.

SKILLS

- **Focus:** GPU high-performance kernel development and LLM training/inference optimization; experience in Attention, MoE routing, quantized GEMM, async state, tensor parallelism, and CUDA Graphs.
- **Development and performance engineering:** C/C++, Python, CUDA, Triton, and PyTorch Extensions; profiling and correctness validation with Nsight Systems, Nsight Compute, and Compute Sanitizer.
- **Languages and credentials:** IELTS 6.5; CET-6; Huawei HCIA-AI certified. Experienced in reading technical documentation, source analysis, and cross-repository debugging.